USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 15 LAB - mPSK
Lab Packet
Spring 2027

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 6.4: I can implement an mPSK modulator and demodulator (BPSK, QPSK, 8PSK), recover carrier and symbol timing, and verify constellation behavior against theory.

This packet is the L15 Lab, in three parts, and you turn it in once, complete, before Lesson 16. Do **Part 1** on your own before the lab period: run the link flowgraph and make the predictions you will check at the bench. **Part 2** is your own measurements, recorded in class: Questions 3 to 6 on your own laptop, and Question 7 at a PlutoSDR station with your team. Do **Part 3** on your own after the lab: compare the measurements with the predictions and answer the analysis questions.

Grading. You submit the packet once, we grade it against Objective 6.4 on the course's 1 / 0 scale, and you may revise it under course policy. Paste screenshots where a question asks for one.

Transmitting. In class we transmit only into a cable, into a shielded enclosure, or at very low power in an ISM band, and only when the lesson says to. Questions 1 to 6 run on your laptop and transmit nothing. Question 7 transmits from a PlutoSDR's TX port, through a cable with an inline 30 dB SMA attenuator, to the same Pluto's RX port, at 915 MHz. Connect the cable and attenuator before you run the flowgraph, and never connect an antenna in this lab. Never set TX attenuation below 30 dB or change the slider's range; the RX input is rated to only a few dBm. We have run the PlutoSDR flowgraph only in simulation, so Question 7 is its first run on hardware.

The flowgraphs are on the L15 lesson page, `L15-psk-tx.grc` and `L15-psk-link.grc`, and on the L15 Lab page, `L15-psk-pluto.grc`. Every number below assumes their settings: 4,000 symbols per second, RRC pulses with $\alpha = 0.35$ at 8 samples per symbol, Gray-coded points, a Costas Loop bandwidth of 0.02, and the default message. Lesson 3 gives the bit error rate with ideal synchronization:

$$P_b^{\text{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right), \qquad P_b^{\text{8PSK}} \approx \frac{2}{3}\,Q\left(\sqrt{\frac{6E_b}{N_0}}\,\sin\frac{\pi}{8}\right).$$

Part 1: Pre-Lab (individual, before the lab period)

1. **Run the link.** Open `L15-psk-link.grc` in GRC and run it with every slider at its default.

   (a) After about 15 s, record the Received label and the Lines label.

   (b) Type 50 into the frequency offset box, wait about 5 s, and paste a screenshot of the running flowgraph's window.

2. Predictions. You will check each of these at the bench.

(a) A faster version of the link sends 6,000 symbols per second. Find the bit rate of BPSK, QPSK, and 8PSK, and the E_s/N_0 each has at $E_b/N_0 = 9$ dB.

	BPSK	QPSK	8PSK
Bit rate			
E_s/N_0 at 9 dB			

(b) Predict the bit error rate from Lesson 3 at each setting you will measure in Question 4. Differential coding roughly doubles it; predict that too.

	QPSK			8PSK		
E_b/N_0 (dB)	5	7	9	9	11	13
Theory						
Differential						

(c) With differential coding off and no frequency offset, you will step the channel phase through 0, 20, 40, ..., 340 degrees, 18 settings, jumping to each from 0. The loop locks to the rotation nearest the received phase. At how many settings should the text arrive for QPSK? For 8PSK?

QPSK: ____

8PSK: ____

(d) Lesson 15 measured the pull-in range of BPSK and QPSK at 12 dB. Estimate the 8PSK loop's pull-in range from that trend, and explain your estimate.

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Part 2: Measurements (in class)

3. **Build a carrier-recovery loop.** Start GRC with an empty canvas. Do not open the lesson's flowgraphs: build the 8PSK carrier-recovery flowgraph yourself as the lab page lists, with a 20 Hz frequency offset.

   (a) Paste a screenshot of your flowgraph on the GRC canvas. Every connection must be black, not red.

   (b) Paste a screenshot of the running window, and record the loop's frequency reading at offsets of 20 Hz and -60 Hz.

   20 Hz:

   -60 Hz:

   (c) Change the Costas Loop's order from 8 to 2 and run it again. Describe the output and the frequency reading, and explain why the loop needs the right order.

4. Bit error rate against theory. Follow the lab page's steps for `L15-psk-link.grc`, with differential coding on, no frequency offset, and the channel phase at 0. At each setting, type the E_b/N_0 into its box, wait 60 s, and record the BER and the number of bits it counted.

(a) Fill in the table, and compare each measurement with the differential prediction in Question 2(b).

	QPSK			8PSK		
E_b/N_0 (dB)	5	7	9	9	11	13
BER						
Bits counted						

(b) Stop the flowgraph, set `M` back to 4 and `diff` to 0, and run it again. Set E_b/N_0 to 7 dB. Record the BER after 60 s and compare it with the theory value in Question 2(b).

(c) Paste a screenshot of the window for 8PSK at 11 dB, and mark one wrong character in the Received label.

5. Carrier recovery. Follow the lab page's steps at $E_b/N_0 = 12$ dB with differential coding on. Before each offset, type 0 into the frequency offset box, wait until lines arrive, and then type the new offset in one jump. Watch the frequency estimate for up to 15 s.

(a) For QPSK, record whether the loop locks, about how long it takes, and the frequency it settles at.

Offset (Hz)	250	350	425
Locks?			
Time			
Settles at (Hz)			

(b) Set `M` to 8 and run the flowgraph again. Record the same for 8PSK, and find its pull-in range.

Offset (Hz)	100	150	200	250
Locks?				
Time				

6. Phase ambiguity. Follow the lab page's steps: set `diff` to 0 and run the flowgraph again, so that the loop locks with no frequency offset and the channel phase at 0, then set $E_b/N_0 = 12$ dB. For each channel phase, type 0 first, wait until lines arrive, and then type the phase in one jump. Mark each setting at which lines still arrive.

(a) Mark the settings that deliver text, and count them.

Phase	0	20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340
QPSK																		
8PSK																		

(b) Set `diff` to 1, run the flowgraph again, and repeat three settings that failed. Record what happens. Then, for QPSK with differential coding off, explain what changes if you step the phase by 20 degrees at a time without returning to 0.

7. QPSK on a PlutoSDR. At a PlutoSDR station, with your team, follow the lab page's steps: connect the Pluto's TX port to its RX port through the 30 dB attenuator, check the Pluto with `iio_info -s`, and run `L15-psk-pluto.grc`. When you finish, stop the flowgraph and unplug the Pluto before you disconnect the cable.

(a) Once lines arrive, paste a screenshot that shows both constellations, the frequency estimate, and the Lines label.

(b) With `diff` = 1, stop and start the flowgraph three times. Then set `diff` to 0 and do it eight more times. Record which runs deliver text.

Run	1	2	3	1	2	3	4	5	6	7	8
Differential	on	on	on	off	off	off	off	off	off	off	off
Text?											

(c) With `diff` = 1, record the BER after 30 s at each TX attenuation, and explain what limits the link at the highest attenuation.

TX attenuation (dB)	60	70	80	89
BER				

Part 3: Writeup (individual, after the lab)

8. Analysis.

(a) Compare your BER table with Question 2(b). Estimate how many dB of E_b/N_0 the link loses to its loops and how many to differential coding, and state the E_b/N_0 a Lesson 2 link budget should deliver for QPSK at a BER of 10^{-4}.

(b) Compare the 8PSK pull-in range with your estimate in Question 2(d), explain why it is smaller than QPSK's, and say what a link between two different Plutos, each accurate to 25 ppm at 915 MHz, would need before its Costas Loop.

(c) Compare Question 6 with Question 2(c) and Question 7(b). Explain why the working settings were where they were, why the Pluto runs behaved as they did, and why a real mPSK link uses differential coding even though it raises the BER.

Documentation: _____